Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Toste Dairy

Physical address of dairy:

609 Santa Fe Grade	Newman	Merced	95360
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1965

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

0054-0090-0001-0000

B. OPERATORS

Toste, John

Operator name: Toste, John

Telephone no.: (209) 862-1812 (Landline) (Cellular)

609 Santa Fe Grade RD	Newman	CA	95360
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

Toste, Sandra

Operator name: Toste, Sandra

Telephone no.: (209) 862-1812 (Landline) (Cellular)

609 Santa Fe Grade RD	Newman	CA	95360
Mailing Address Number and Street	City	State	Zip Code

C. OWNERS

Toste, John

Legal owner name: Toste, John

Telephone no.: (209) 862-1812

Landline	Cellular

609 Santa Fe Grade RD	Newman	CA	95360
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

Toste, Sandra

Legal owner name: Toste, Sandra

Telephone no.: (209) 862-1812

Landline	Cellular

609 Santa Fe Grade RD	Newman	CA	95360
Mailing Address Number and Street	City	State	Zip Code

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	170	175	750	250	0
Number under roof	1,450	0	0	0	0	0
Maximum number	1,450	170	175	750	250	0
Average number	1,450	170	175	750	250	0
Avg live weight (lbs)	1,400	1,400	900	600		

Predominant milk cow breed: Holstein

Average milk production: 72 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 49,480.19 *tons per reporting period*

Total nitrogen from manure: 622,333.58 *lbs per reporting period*

After ammonia losses (30% loss applied): 435,633.51 *lbs per reporting period*

Total phosphorus from manure: 103,000.25 *lbs per reporting period*

Total potassium from manure: 279,593.68 *lbs per reporting period*

Total salt from manure: 721,824.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 10,484,000 *gallons*

Total nitrogen generated: 70,726.14 *lbs*

Total phosphorus generated: 5,905.86 *lbs*

Total potassium generated: 71,304.02 *lbs*

Total salt generated: 619,433.33 *lbs*

	10,484,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	10,484,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
08/09/2023	Solid commercial fertilizer *UN 32*	29.43 *ton*	As-is	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	18,835.20	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	18,835.20	0.00	0.00	0.00

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	Separator solids	2,200.00 *ton*	Dry-weight	72.4		12,600.00	10,000.00	2,100.00		30.10
04/21/2023	Corral solids	720.00 *ton*	Dry-weight	61.6		26,600.00	12,200.00	66,400.00		23.70

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	30,010.18	18,890.11	39,266.78	496,585.92
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	30,010.18	18,890.11	39,266.78	496,585.92

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Azevedo Home	18	18	2	both	X054-X130-X005-XXXX
Azevedo/Botelho	44	44	2	process wastewater	X054-X090-X004-XXXX X054-X100-X030-XXXX
Botelho Back 40	36	36	2	both	X054-X100-X030-XXXX
Botelho Pasture 40	21	21	2	both	X054-X100-X030-XXXX
Costa	83	83	2	both	X054-X100-X018-XXXX
Creamery	37	37	2	manure	X026-X016-X032-XXXX X026-X016-X033-XXXX X054-X050-X004-XXXX
Home 1 & 2	88	88	2	both	X054-X090-X001-XXXX X054-X100-X023-XXXX
New Pasture	5	5	0	none	X054-X050-X015-XXXX
Totals for areas that were used for application	327	327	14		
Totals for areas that were not used for application	5	5	0		
Land application area totals	332	332	14		

B. CROPS AND HARVESTS

Azevedo Home

Field name: Azevedo Home

10/12/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 18 Plant date: 10/12/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/20/2023	272.00 *ton*	As-is		72.2	5,800.00	900.00	8,300.00		11.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.11	175.29	27.20	250.84	999.81

Azevedo Home

05/31/2023: Corn, silage

Crop: Corn, silage Acres planted: 18 Plant date: 05/31/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/26/2023	451.00 *ton*	As-is		70.8	3,740.00	660.00	6,320.00		7.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	25.06	187.42	33.07	316.70	1,038.90

Azevedo/Botelho

Field name: Azevedo/Botelho

10/11/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 44 Plant date: 10/11/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/20/2023	620.00 *ton*	As-is		54.0	7,100.00	1,400.00	10,700.00		10.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	14.09	200.09	39.45	301.55	1,374.15

06/27/2023: Corn, silage

Crop: Corn, silage Acres planted: 44 Plant date: 06/27/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/19/2023	1,100.00 *ton*	As-is		63.4	4,030.00	970.00	5,370.00		6.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	25.00	201.50	48.50	268.50	1,152.90

Botelho Back 40

Field name: Botelho Back 40

10/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 36 Plant date: 10/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/20/2023	480.00 *ton*	As-is		70.8	7,300.00	1,200.00	12,500.00		12.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	13.33	194.67	32.00	333.33	949.97

06/29/2023: Corn, silage

Crop: Corn, silage Acres planted: 36 Plant date: 06/29/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/19/2023	898.00 *ton*	As-is		64.6	4,250.00	910.00	5,160.00		6.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	24.94	212.03	45.40	257.43	1,130.28

Botelho Pasture 40

Field name: Botelho Pasture 40

10/12/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 21 Plant date: 10/12/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/20/2023	320.00 *ton*	As-is		72.2	5,800.00	900.00	8,300.00		11.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.24	176.76	27.43	252.95	1,008.21

Botelho Pasture 40

05/31/2023: Corn, silage

Crop: Corn, silage Acres planted: 21 Plant date: 05/31/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/26/2023	514.00 *ton*	As-is		64.8	4,010.00	910.00	5,920.00		4.49

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	24.48	196.30	44.55	289.80	773.68

Costa

Field name: Costa

10/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 83 Plant date: 10/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	1,164.00 *ton*	As-is		66.4	6,600.00	1,100.00	8,600.00		8.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	14.02	185.12	30.85	241.21	782.21

06/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 83 Plant date: 06/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/19/2023	1,854.00 *ton*	As-is		60.8	4,430.00	710.00	5,140.00		5.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	22.34	197.91	31.72	229.63	1,015.72

Creamery

Field name: Creamery

10/06/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 37 Plant date: 10/06/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	590.00 *ton*	As-is		54.0	7,100.00	1,400.00	10,700.00		10.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.95	226.43	44.65	341.24	1,555.05

06/07/2023: Corn, silage

Crop: Corn, silage Acres planted: 37 Plant date: 06/07/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/26/2023	941.00 *ton*	As-is		70.0	3,120.00	490.00	3,470.00		5.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	25.43	158.70	24.92	176.50	808.75

Home 1 & 2

Field name: Home 1 & 2

10/07/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 88 Plant date: 10/07/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/20/2023	1,257.00 *ton*	As-is		54.0	7,100.00	1,400.00	10,700.00		10.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	14.28	202.83	40.00	305.68	1,392.98

Home 1 & 2

06/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 88 Plant date: 06/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/25/2023	1,801.00 *ton*	As-is		68.6	3,930.00	680.00	4,400.00		4.83

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	20.47	160.86	27.83	180.10	620.78

NUTRIENT BUDGET

A. LAND APPLICATIONS

Azevedo Home - 10/12/2022: Oats, silage-soft dough

Field name: Azevedo Home

Crop: Oats, silage-soft dough — Plant date: 10/12/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	120.03	16.29	138.86	1,233.72	288,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	256.11	3,564,000.00 *gal*
Application event totals		120.03	16.29	138.86	1,489.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	115.18	11.84	133.88	1,256.87	384,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	124.17	1,728,000.00 *gal*
Application event totals		115.18	11.84	133.88	1,381.04	

Azevedo Home - 05/31/2023: Corn, silage

Field name: Azevedo Home

Crop: Corn, silage — Plant date: 05/31/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	171.56	136.16	28.59	4,098.42	444.00 *ton*
Application event totals		171.56	136.16	28.59	4,098.42	

Azevedo Home - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	279.39	3,888,000.00 *gal*
Application event totals		0.00	0.00	0.00	279.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	186.26	2,592,000.00 *gal*
Application event totals		0.00	0.00	0.00	186.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	186.26	2,592,000.00 *gal*
Application event totals		0.00	0.00	0.00	186.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	279.39	3,888,000.00 *gal*
Application event totals		0.00	0.00	0.00	279.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	57.60	0.00	0.00	0.00	
Application event totals		57.60	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	139.70	1,944,000.00 *gal*
Application event totals		0.00	0.00	0.00	139.70	

Azevedo Home - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	139.70	1,944,000.00 *gal*
Application event totals		0.00	0.00	0.00	139.70	

09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	186.26	2,592,000.00 *gal*
Application event totals		0.00	0.00	0.00	186.26	

Azevedo/Botelho - 10/11/2022: Oats, silage-soft dough

Field name: Azevedo/Botelho

Crop: Oats, silage-soft dough

Plant date: 10/11/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	81.84	11.11	94.68	841.18	480,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	95.25	3,240,000.00 *gal*
Application event totals		81.84	11.11	94.68	936.42	

01/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	137.65	10.49	111.43	694.79	576,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	166.68	5,670,000.00 *gal*
Application event totals		137.65	10.49	111.43	861.47	

Azevedo/Botelho - 06/27/2023: Corn, silage

Azevedo/Botelho - 06/27/2023: Corn, silage

Field name: Azevedo/Botelho

Crop: Corn, silage

Plant date: 06/27/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	178.96	6.37	60.01	262.34	280,000.00 *gal*
Application event totals		178.96	6.37	60.01	262.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	190.49	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	190.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	342.89	11,664,000.00 *gal*
Application event totals		0.00	0.00	0.00	342.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	57.60	0.00	0.00	0.00	
Application event totals		57.60	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	190.49	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	190.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	190.49	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	190.49	

Azevedo/Botelho - 06/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	38.36	3.23	49.20	449.90	336,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	342.89	11,664,000.00 *gal*
Application event totals		38.36	3.23	49.20	792.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	152.39	5,184,000.00 *gal*
Application event totals		0.00	0.00	0.00	152.39	

Botelho Back 40 - 10/08/2022: Oats, silage-soft dough

Field name: Botelho Back 40

Crop: Oats, silage-soft dough

Plant date: 10/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/16/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	239.09	83.23	45.21	2,162.40	680.00 *ton*
Application event totals		239.09	83.23	45.21	2,162.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/13/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	349.24	9,720,000.00 *gal*
Application event totals		0.00	0.00	0.00	349.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	93.13	2,592,000.00 *gal*
Application event totals		0.00	0.00	0.00	93.13	

Botelho Back 40 - 06/29/2023: Corn, silage

Field name: Botelho Back 40

Crop: Corn, silage

Plant date: 06/29/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	171.86	6.12	57.63	251.93	220,000.00 *gal*
Application event totals		171.86	6.12	57.63	251.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	413.91	11,520,000.00 *gal*
Application event totals		0.00	0.00	0.00	413.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	279.39	7,776,000.00 *gal*
Application event totals		0.00	0.00	0.00	279.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	57.60	0.00	0.00	0.00	
Application event totals		57.60	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	186.26	5,184,000.00 *gal*
Application event totals		0.00	0.00	0.00	186.26	

Botelho Back 40 - 06/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	310.43	8,640,000.00 *gal*
Application event totals		0.00	0.00	0.00	310.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	40.19	3.38	51.54	471.33	288,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	206.96	5,760,000.00 *gal*
Application event totals		40.19	3.38	51.54	678.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	206.96	5,760,000.00 *gal*
Application event totals		0.00	0.00	0.00	206.96	

Botelho Pasture 40 - 10/12/2022: Oats, silage-soft dough

Field name: Botelho Pasture 40

Crop: Oats, silage-soft dough

Plant date: 10/12/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	232.15	92.02	330.92	2,346.85	320.00 *ton*
Application event totals		232.15	92.02	330.92	2,346.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	359.22	5,832,000.00 *gal*
Application event totals		0.00	0.00	0.00	359.22	

Botelho Pasture 40 - 05/31/2023: Corn, silage

Field name: Botelho Pasture 40

Crop: Corn, silage

Plant date: 05/31/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	98.73	10.15	114.75	1,077.32	384,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	239.48	3,888,000.00 *gal*
Application event totals		98.73	10.15	114.75	1,316.79	

07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	239.48	3,888,000.00 *gal*
Application event totals		0.00	0.00	0.00	239.48	

07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	103.34	8.70	132.53	1,211.98	432,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	239.48	3,888,000.00 *gal*
Application event totals		103.34	8.70	132.53	1,451.46	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	239.48	3,888,000.00 *gal*
Application event totals		0.00	0.00	0.00	239.48	

08/09/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	57.60	0.00	0.00	0.00	
Application event totals		57.60	0.00	0.00	0.00	

Botelho Pasture 40 - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	239.48	3,888,000.00 *gal*
Application event totals		0.00	0.00	0.00	239.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	119.74	1,944,000.00 *gal*
Application event totals		0.00	0.00	0.00	119.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	319.30	5,184,000.00 *gal*
Application event totals		0.00	0.00	0.00	319.30	

Costa - 10/10/2022: Oats, silage-soft dough

Field name: Costa

Crop: Oats, silage-soft dough

Plant date: 10/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	231.28	91.67	329.68	2,338.01	1,260.00 *ton*
Application event totals		231.28	91.67	329.68	2,338.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/02/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	201.97	12,960,000.00 *gal*
Application event totals		0.00	0.00	0.00	201.97	

Costa - 06/28/2023: Corn, silage

Field name: Costa

Crop: Corn, silage

Plant date: 06/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	87.43	8.99	101.62	954.01	1,344,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	424.14	27,216,000.00 *gal*
Application event totals		87.43	8.99	101.62	1,378.14	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	424.14	27,216,000.00 *gal*
Application event totals		0.00	0.00	0.00	424.14	

08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	69.73	5.87	89.42	817.72	1,152,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	181.77	11,664,000.00 *gal*
Application event totals		69.73	5.87	89.42	999.49	

08/23/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	57.60	0.00	0.00	0.00	
Application event totals		57.60	0.00	0.00	0.00	

08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	181.77	11,664,000.00 *gal*
Application event totals		0.00	0.00	0.00	181.77	

Costa - 06/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	63.92	5.38	81.97	749.58	1,056,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	242.36	15,552,000.00 *gal*
Application event totals		63.92	5.38	81.97	991.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	151.48	9,720,000.00 *gal*
Application event totals		0.00	0.00	0.00	151.48	

Creamery - 10/06/2022: Oats, silage-soft dough

Field name: Creamery

Crop: Oats, silage-soft dough Plant date: 10/06/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/16/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	239.47	83.36	45.28	2,165.84	700.00 *ton*
Application event totals		239.47	83.36	45.28	2,165.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	339.80	9,720,000.00 *gal*
Application event totals		0.00	0.00	0.00	339.80	

Creamery - 06/07/2023: Corn, silage

Field name: Creamery

Crop: Corn, silage Plant date: 06/07/2023

Creamery - 06/07/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	152.26	120.84	25.38	3,637.38	810.00 *ton*
Application event totals		152.26	120.84	25.38	3,637.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	396.43	11,340,000.00 *gal*
Application event totals		0.00	0.00	0.00	396.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	226.53	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	226.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	169.90	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	169.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	226.53	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	226.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	57.60	0.00	0.00	0.00	
Application event totals		57.60	0.00	0.00	0.00	

Creamery - 06/07/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	169.90	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	169.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	226.53	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	226.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	226.53	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	226.53	

Home 1 & 2 - 10/07/2022: Oats, silage-soft dough

Field name: Home 1 & 2

Crop: Oats, silage-soft dough

Plant date: 10/07/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	234.74	81.72	44.38	2,123.08	1,632.00 *ton*
Application event totals		234.74	81.72	44.38	2,123.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/16/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	190.49	12,960,000.00 *gal*
Application event totals		0.00	0.00	0.00	190.49	

Home 1 & 2 - 10/07/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	99.22	6,750,000.00 *gal*
Application event totals		0.00	0.00	0.00	99.22	

Home 1 & 2 - 06/08/2023: Corn, silage

Field name: Home 1 & 2

Crop: Corn, silage

Plant date: 06/08/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	106.02	10.90	123.23	1,156.89	1,728,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	400.04	27,216,000.00 *gal*
Application event totals		106.02	10.90	123.23	1,556.93	

07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	285.74	19,440,000.00 *gal*
Application event totals		0.00	0.00	0.00	285.74	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	214.31	14,580,000.00 *gal*
Application event totals		0.00	0.00	0.00	214.31	

08/10/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	57.60	0.00	0.00	0.00	
Application event totals		57.60	0.00	0.00	0.00	

Home 1 & 2 - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	238.12	16,200,000.00 *gal*
Application event totals		0.00	0.00	0.00	238.12	

08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	190.49	12,960,000.00 *gal*
Application event totals		0.00	0.00	0.00	190.49	

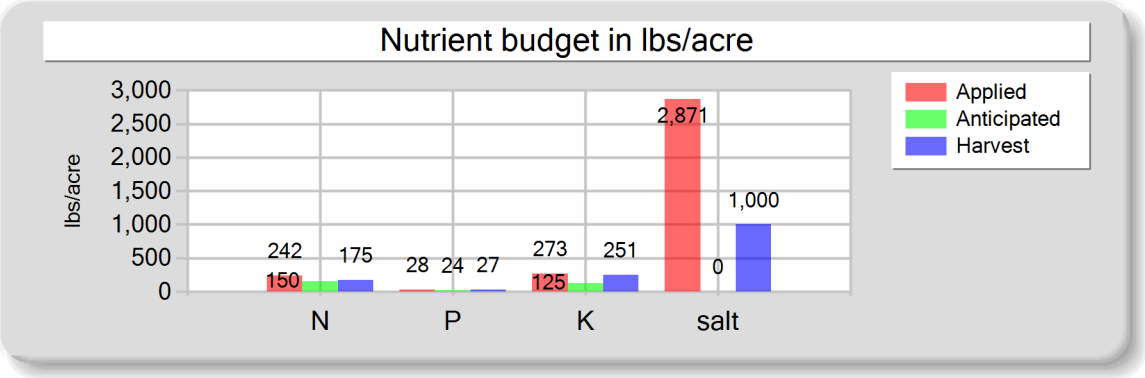
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	87.69	7.38	112.45	1,028.35	1,536,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	142.87	9,720,000.00 *gal*
Application event totals		87.69	7.38	112.45	1,171.22	

B. NUTRIENT BUDGET

Azevedo Home - 10/12/2022: Oats, silage-soft dough

Field name: Azevedo Home | Crop: Oats, silage-soft dough | Plant date: 10/12/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	235.22	28.13	272.74	2,490.59
Fresh water	0.00	0.00	0.00	380.28
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	242.22	28.13	272.74	2,870.87
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	175.29	27.20	250.84	999.81
Nutrient balance	66.93	0.93	21.89	1,871.06
Applied to removed ratio	1.38	1.03	1.09	2.87

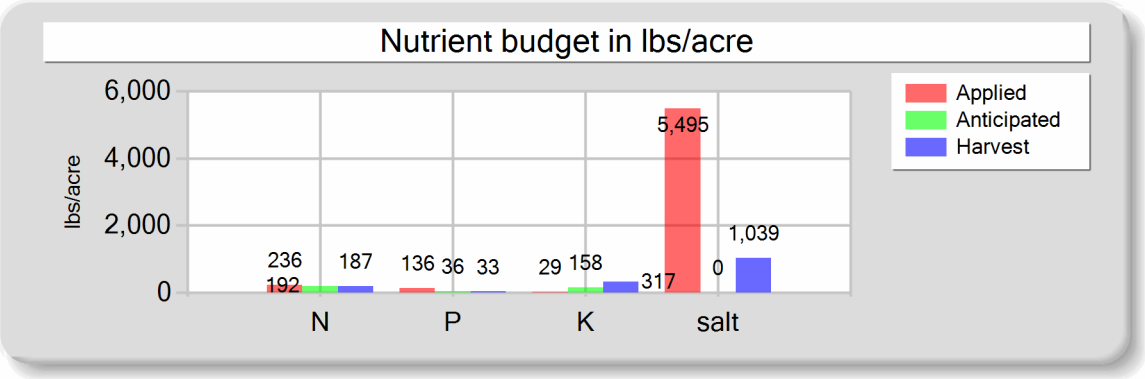
Fresh water applied	
5,292,000.00	*gallons*
194.89	*acre-inches*
10.83	*inches/acre*

Process wastewater applied	
672,000.00	*gallons*
24.75	*acre-inches*
1.37	*inches/acre*

Total harvests for the crop	
1	*harvests*

Azevedo Home - 05/31/2023: Corn, silage

Field name: Azevedo Home | Crop: Corn, silage | Plant date: 05/31/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	57.60	0.00	0.00	0.00
Dry manure	171.56	136.16	28.59	4,098.42
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,396.95
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	236.16	136.16	28.59	5,495.37
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	187.42	33.07	316.70	1,038.90
Nutrient balance	48.75	103.09	-288.11	4,456.47
Applied to removed ratio	1.26	4.12	0.09	5.29

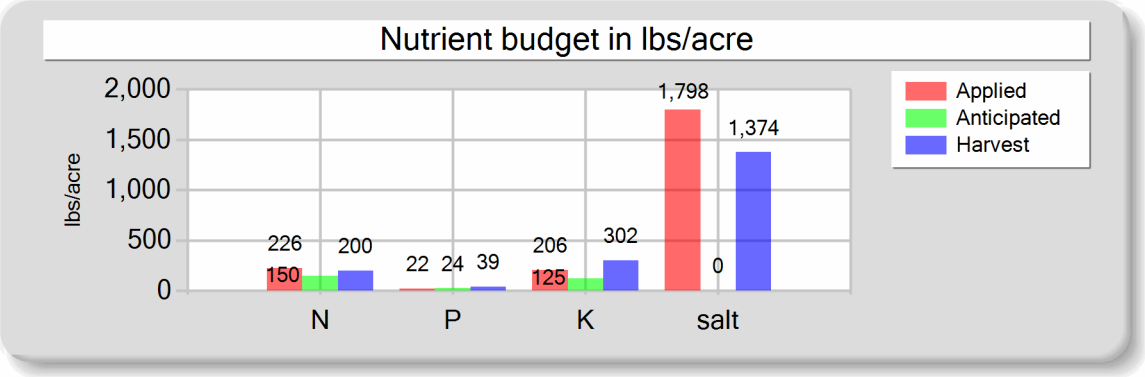
Fresh water applied	
19,440,000.00	*gallons*
715.91	*acre-inches*
39.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Azevedo/Botelho - 10/11/2022: Oats, silage-soft dough

Field name: Azevedo/Botelho Crop: Oats, silage-soft dough Plant date: 10/11/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	219.49	21.59	206.11	1,535.97
Fresh water	0.00	0.00	0.00	261.93
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	226.49	21.59	206.11	1,797.89
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	200.09	39.45	301.55	1,374.15
Nutrient balance	26.40	-17.86	-95.44	423.75
Applied to removed ratio	1.13	0.55	0.68	1.31

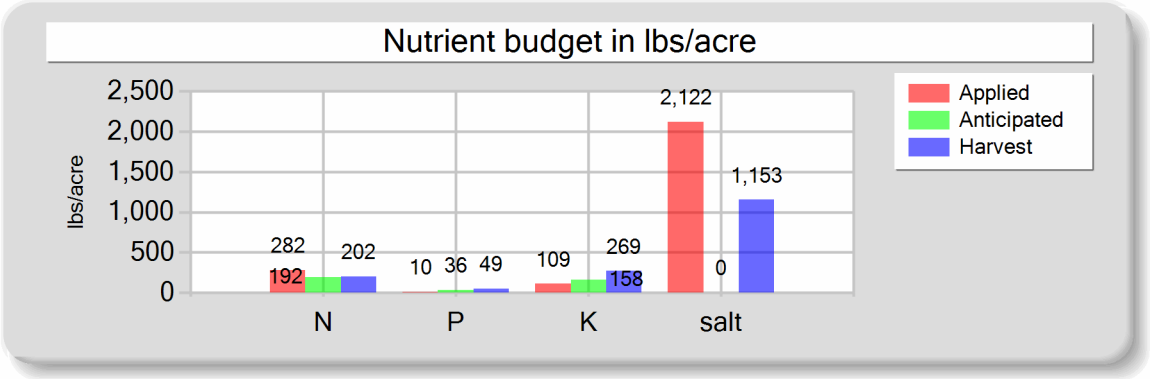
Fresh water applied	
8,910,000.00	*gallons*
328.13	*acre-inches*
7.46	*inches/acre*

Process wastewater applied	
1,056,000.00	*gallons*
38.89	*acre-inches*
0.88	*inches/acre*

Total harvests for the crop	
1	*harvests*

Azevedo/Botelho - 06/27/2023: Corn, silage

Field name: Azevedo/Botelho | Crop: Corn, silage | Plant date: 06/27/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	57.60	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	217.33	9.60	109.20	712.24
Fresh water	0.00	0.00	0.00	1,409.65
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	281.93	9.60	109.20	2,121.89
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	201.50	48.50	268.50	1,152.90
Nutrient balance	80.43	-38.90	-159.30	968.99
Applied to removed ratio	1.40	0.20	0.41	1.84

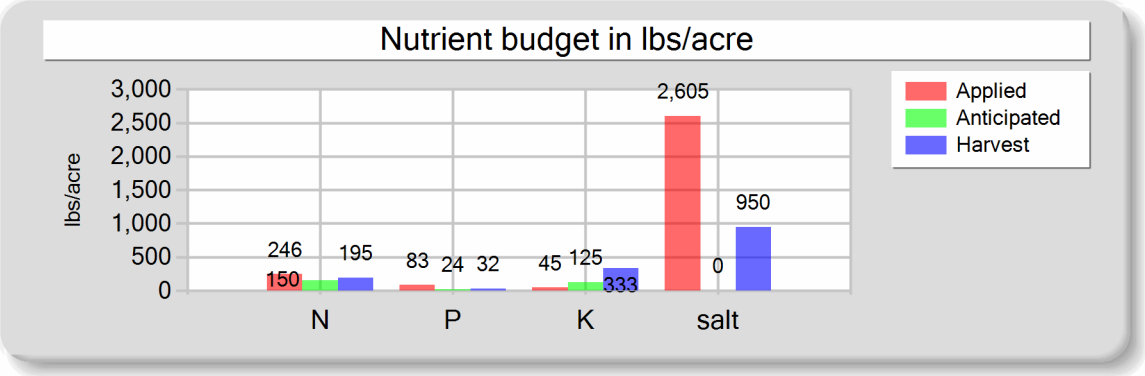
Fresh water applied	
47,952,000.00	*gallons*
1,765.91	*acre-inches*
40.13	*inches/acre*

Process wastewater applied	
616,000.00	*gallons*
22.69	*acre-inches*
0.52	*inches/acre*

Total harvests for the crop	
1	*harvests*

Botelho Back 40 - 10/08/2022: Oats, silage-soft dough

Field name: Botelho Back 40 Crop: Oats, silage-soft dough Plant date: 10/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	239.09	83.23	45.21	2,162.40
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	442.37
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	246.09	83.23	45.21	2,604.77
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	194.67	32.00	333.33	949.97
Nutrient balance	51.42	51.23	-288.13	1,654.80
Applied to removed ratio	1.26	2.60	0.14	2.74

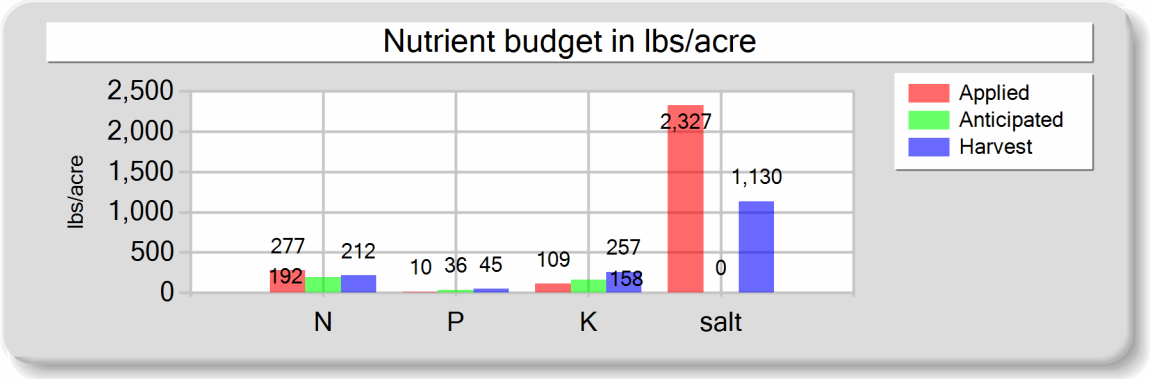
Fresh water applied	
12,312,000.00	*gallons*
453.41	*acre-inches*
12.59	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Botelho Back 40 - 06/29/2023: Corn, silage

Field name: Botelho Back 40 Crop: Corn, silage Plant date: 06/29/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	57.60	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	212.05	9.50	109.17	723.25
Fresh water	0.00	0.00	0.00	1,603.91
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	276.65	9.50	109.17	2,327.16
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	212.03	45.40	257.43	1,130.28
Nutrient balance	64.62	-35.89	-148.26	1,196.88
Applied to removed ratio	1.30	0.21	0.42	2.06

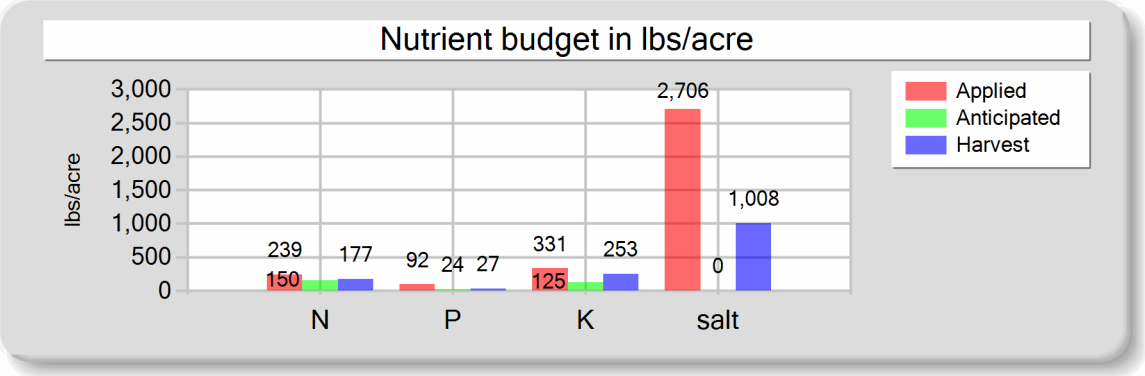
Fresh water applied	
44,640,000.00	*gallons*
1,643.94	*acre-inches*
45.66	*inches/acre*

Process wastewater applied	
508,000.00	*gallons*
18.71	*acre-inches*
0.52	*inches/acre*

Total harvests for the crop	
1	*harvests*

Botelho Pasture 40 - 10/12/2022: Oats, silage-soft dough

Field name: Botelho Pasture 40 Crop: Oats, silage-soft dough Plant date: 10/12/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	232.15	92.02	330.92	2,346.85
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	359.22
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	239.15	92.02	330.92	2,706.07
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	176.76	27.43	252.95	1,008.21
Nutrient balance	62.39	64.59	77.97	1,697.85
Applied to removed ratio	1.35	3.35	1.31	2.68

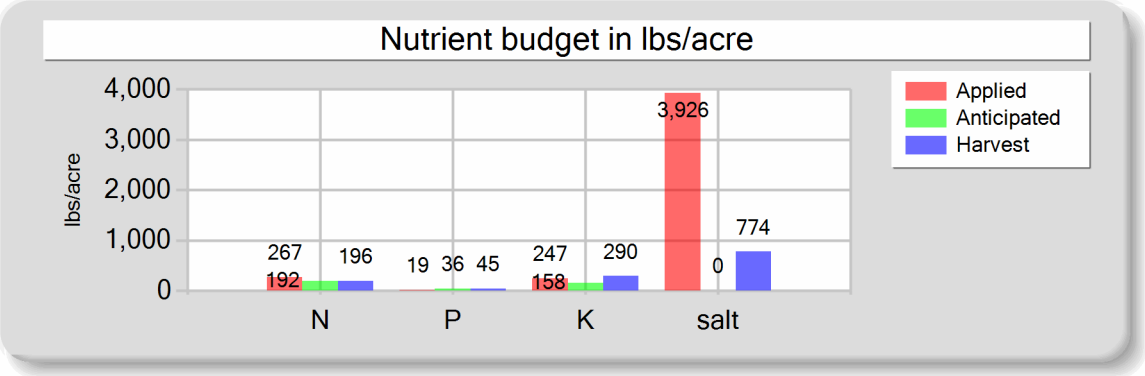
Fresh water applied	
5,832,000.00	*gallons*
214.77	*acre-inches*
10.23	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Botelho Pasture 40 - 05/31/2023: Corn, silage

Field name: Botelho Pasture 40 Crop: Corn, silage Plant date: 05/31/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	57.60	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	202.07	18.85	247.28	2,289.30
Fresh water	0.00	0.00	0.00	1,636.43
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	266.67	18.85	247.28	3,925.73
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	196.30	44.55	289.80	773.68
Nutrient balance	70.37	-25.70	-42.52	3,152.04
Applied to removed ratio	1.36	0.42	0.85	5.07

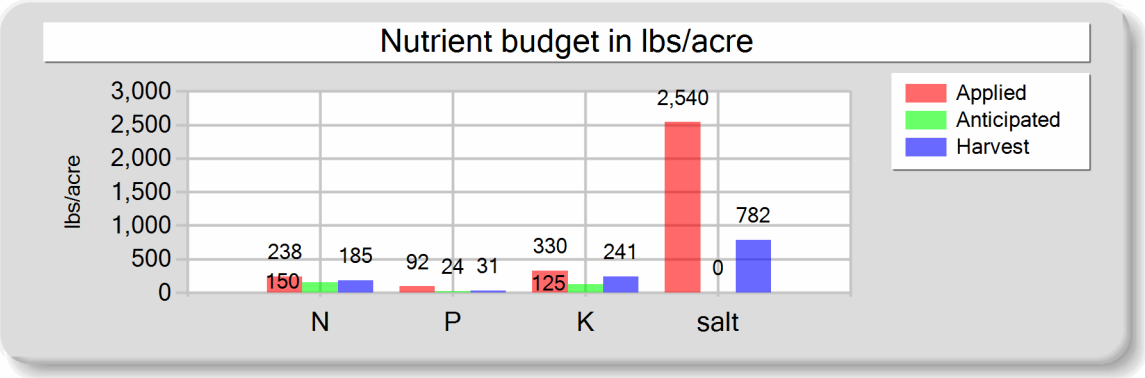
Fresh water applied	
26,568,000.00	*gallons*
978.41	*acre-inches*
46.59	*inches/acre*

Process wastewater applied	
816,000.00	*gallons*
30.05	*acre-inches*
1.43	*inches/acre*

Total harvests for the crop	
1	*harvests*

Costa - 10/10/2022: Oats, silage-soft dough

Field name: Costa | Crop: Oats, silage-soft dough | Plant date: 10/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	231.28	91.67	329.68	2,338.01
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	201.97
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	238.28	91.67	329.68	2,539.98
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	185.12	30.85	241.21	782.21
Nutrient balance	53.16	60.82	88.46	1,757.77
Applied to removed ratio	1.29	2.97	1.37	3.25

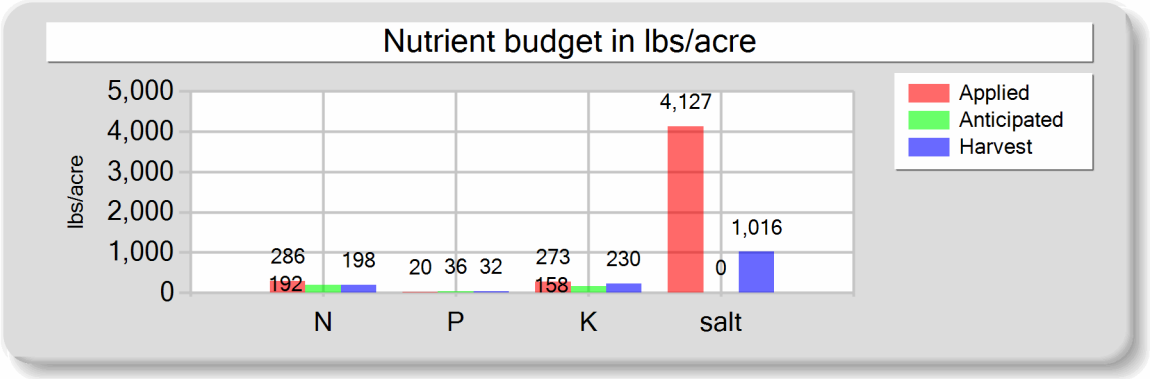
Fresh water applied	
12,960,000.00	*gallons*
477.27	*acre-inches*
5.75	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Costa - 06/28/2023: Corn, silage

Field name: Costa | Crop: Corn, silage | Plant date: 06/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	57.60	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	221.07	20.24	273.00	2,521.31
Fresh water	0.00	0.00	0.00	1,605.65
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	285.67	20.24	273.00	4,126.96
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	197.91	31.72	229.63	1,015.72
Nutrient balance	87.76	-11.48	43.37	3,111.24
Applied to removed ratio	1.44	0.64	1.19	4.06

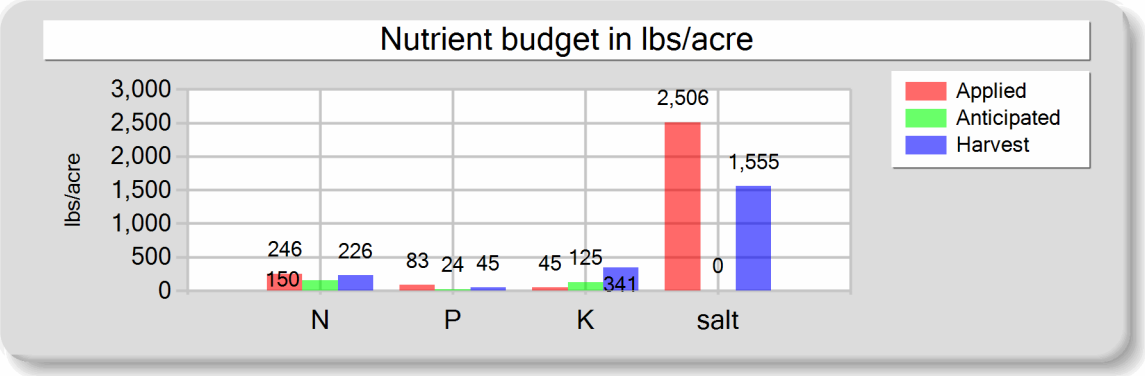
Fresh water applied	
103,032,000.00	*gallons*
3,794.32	*acre-inches*
45.71	*inches/acre*

Process wastewater applied	
3,552,000.00	*gallons*
130.81	*acre-inches*
1.58	*inches/acre*

Total harvests for the crop	
1	*harvests*

Creamery - 10/06/2022: Oats, silage-soft dough

Field name: Creamery | Crop: Oats, silage-soft dough | Plant date: 10/06/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	239.47	83.36	45.28	2,165.84
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	339.80
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	246.47	83.36	45.28	2,505.64
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	226.43	44.65	341.24	1,555.05
Nutrient balance	20.04	38.72	-295.96	950.59
Applied to removed ratio	1.09	1.87	0.13	1.61

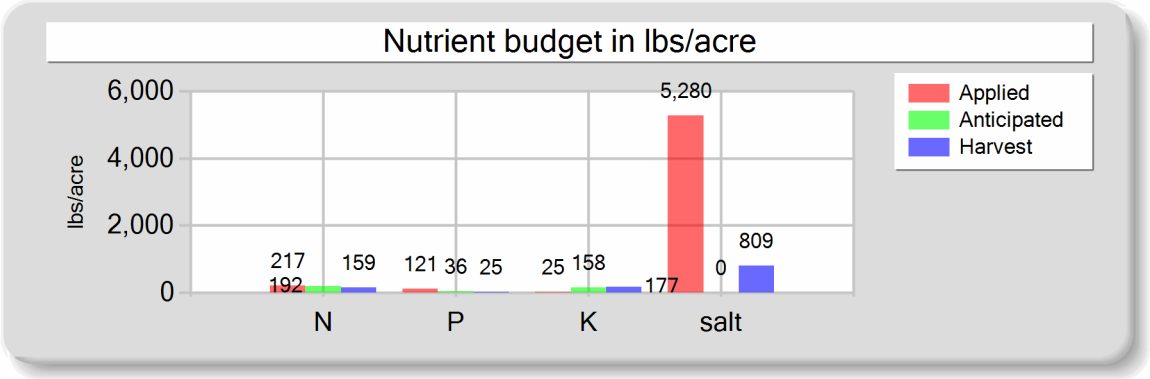
Fresh water applied	
9,720,000.00	*gallons*
357.95	*acre-inches*
9.67	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Creamery - 06/07/2023: Corn, silage

Field name: Creamery | Crop: Corn, silage | Plant date: 06/07/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	57.60	0.00	0.00	0.00
Dry manure	152.26	120.84	25.38	3,637.38
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,642.36
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	216.86	120.84	25.38	5,279.75
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	158.70	24.92	176.50	808.75
Nutrient balance	58.16	95.92	-151.12	4,470.99
Applied to removed ratio	1.37	4.85	0.14	6.53

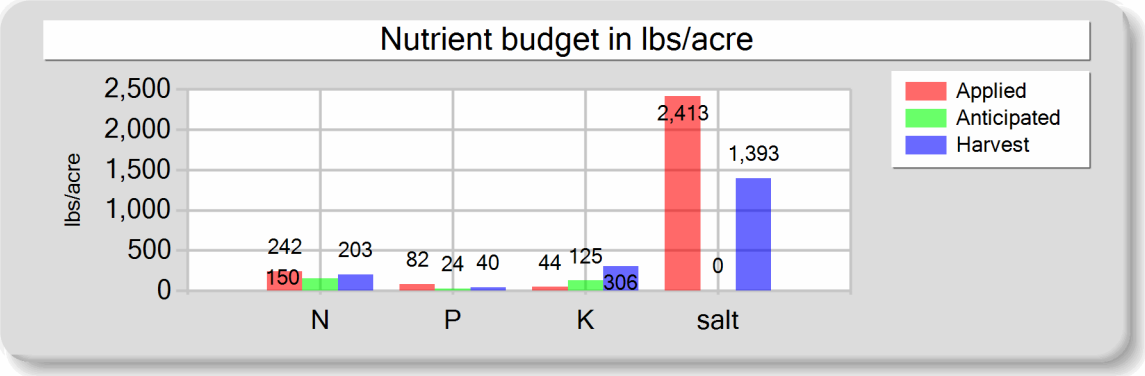
Fresh water applied	
46,980,000.00	*gallons*
1,730.11	*acre-inches*
46.76	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Home 1 & 2 - 10/07/2022: Oats, silage-soft dough

Field name: Home 1 & 2 Crop: Oats, silage-soft dough Plant date: 10/07/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	234.74	81.72	44.38	2,123.08
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	289.71
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	241.74	81.72	44.38	2,412.79
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	202.83	40.00	305.68	1,392.98
Nutrient balance	38.91	41.72	-261.30	1,019.81
Applied to removed ratio	1.19	2.04	0.15	1.73

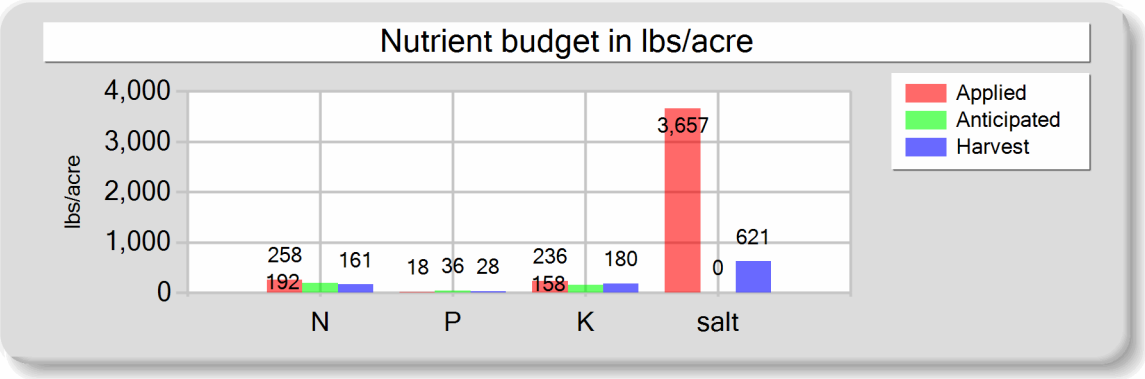
Fresh water applied	
19,710,000.00	*gallons*
725.85	*acre-inches*
8.25	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Home 1 & 2 - 06/08/2023: Corn, silage

Field name: Home 1 & 2 Crop: Corn, silage Plant date: 06/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	57.60	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	193.71	18.28	235.67	2,185.24
Fresh water	0.00	0.00	0.00	1,471.56
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	258.31	18.28	235.67	3,656.80
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	160.86	27.83	180.10	620.78
Nutrient balance	97.45	-9.55	55.57	3,036.02
Applied to removed ratio	1.61	0.66	1.31	5.89

Fresh water applied	
100,116,000.00	*gallons*
3,686.93	*acre-inches*
41.90	*inches/acre*

Process wastewater applied	
3,264,000.00	*gallons*
120.20	*acre-inches*
1.37	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 11/02/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,500.00	10,900.00	39,200.00							27.80
DL	100.00	100.00	100.00							0.01

Settling Basin Solids

Sample and source description: Settling Basin Solids

Sample date: 11/02/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 78.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	29,300.00	10,200.00	5,540.00	30,800.00	8,120.00	2,660.00	10,900.00	1,910.00		26.50
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 04/24/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,600.00	12,200.00	66,400.00	28,500.00	14,600.00	24,500.00	10,100.00	1,120.00		23.70
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

Settling Basin Solids

Sample and source description: Settling Basin Solids

Sample date: 04/24/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,600.00	10,000.00	2,100.00							30.10
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 09/27/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 49.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,800.00	9,000.00	62,800.00							39.80
DL	100.00	100.00	100.00							0.01

Settling Basin Solids

Sample and source description: Settling Basin Solids

Sample date: 10/23/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 85.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,200.00	6,000.00	11,600.00							20.20
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

4QWW

Sample and source description: 4QWW

Sample date: 12/13/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	899.00	518.00	0.00	0.00	122.00	1,040.00								7,560.00	9,240
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

1QWW

Sample and source description: 1QWW

Sample date: 03/31/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,260.00	501.00	0.00	0.00	96.00	1,020.00	350.00	171.00	450.00	5,040.00	0.00	271.00	544.00	7,840.00	6,360
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	0.50	1.00	0.50	1.00	10

Slurry

Sample and source description: Slurry

Sample date: 04/24/2023 Material type: Process wastewater sludge Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3,370.00	980.00	0.00	0.00	120.00	1,130.00								1,430.00	4,940
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

2QWW

Sample and source description: 2QWW

Sample date: 05/18/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	647.00	375.00	0.00	0.00	66.50	752.00								8,050.00	7,060
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

3QWW

Sample and source description: 3QWW

Sample date: 09/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	602.00	409.00	0.00	0.00	50.70	772.00	176.00	178.00	374.00	3,840.00	0.00	276.00	455.00	8,060.00	7,060
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	0.50	1.00	0.50	1.00	10

4QWW

Sample and source description: 4QWW

Sample date: 12/28/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,150.00	529.00	0.00	0.00	90.80	1,350.00								1,150.00	5,760
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

Canal

CCID Canal

Sample description: CCID Canal

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								228.00	155
DL	1.00	0.50	0.10								10.00	10

D. SOIL ANALYSES

Azevedo Home

Azevedo Home

Soil Sample

Sample and source description: Soil Sample

Sample date: 10/05/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			1,268.00				
DL			4.00				

Azevedo/Botelho

Soil Sample

Sample and source description: Soil Sample

Sample date: 10/05/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			1,888.00				
DL			4.00				

Botelho Back 40

Soil Sample

Sample and source description: Soil Sample

Sample date: 11/03/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			2,083.00				
DL			4.00				

Botelho Pasture 40

Botelho Pasture 40

Soil Sample

Sample and source description: Soil Sample

Sample date: 11/03/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			1,607.00				
DL			4.00				

Costa

Soil Sample

Sample and source description: Soil Sample

Sample date: 11/03/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			941.00				
DL			4.00				

Creamery

Soil Sample

Sample and source description: Soil Sample

Sample date: 10/05/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			329.00				
DL			4.00				

Home 1 & 2

Home 1 & 2

Soil Sample

Sample and source description: Soil Sample

Sample date: 10/05/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			1,930.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

Azevedo Home - 10/12/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,500.00	1,380.00	14,400.00		8.96
DL	100.00	100.00	100.00		0.01

Azevedo Home - 05/31/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/27/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,740.00	660.00	6,320.00		7.10
DL	100.00	100.00	100.00		0.01

Azevedo/Botelho - 10/11/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 74.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,600.00	1,110.00	14,100.00		10.00
DL	100.00	100.00	100.00		0.01

Azevedo/Botelho - 06/27/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,030.00	970.00	5,370.00		6.30
DL	100.00	100.00	100.00		0.01

Botelho Back 40 - 10/08/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 79.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,090.00	1,240.00	13,700.00		12.30
DL	100.00	100.00	100.00		0.01

Botelho Back 40 - 06/29/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,250.00	910.00	5,160.00		6.40
DL	100.00	100.00	100.00		0.01

Botelho Pasture 40 - 10/12/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 78.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,600.00	1,300.00	14,500.00		11.70
DL	100.00	100.00	100.00		0.01

Botelho Pasture 40 - 05/31/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/27/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,010.00	910.00	5,920.00		4.49
DL	100.00	100.00	100.00		0.01

Costa - 10/10/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,200.00	1,730.00	20,500.00		10.80
DL	100.00	100.00	100.00		0.01

Costa - 06/28/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,430.00	710.00	5,140.00		5.80
DL	100.00	100.00	100.00		0.01

Creamery - 10/06/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 46.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,900.00	1,300.00	15,100.00		8.80
DL	100.00	100.00	100.00		0.01

Creamery - 06/07/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/27/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,120.00	490.00	3,470.00		5.30
DL	100.00	100.00	100.00		0.01

Home 1 & 2 - 10/07/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 78.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,690.00	1,180.00	12,800.00		10.80
DL	100.00	100.00	100.00		0.01

Home 1 & 2 - 06/08/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/27/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,930.00	680.00	4,400.00		4.83
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

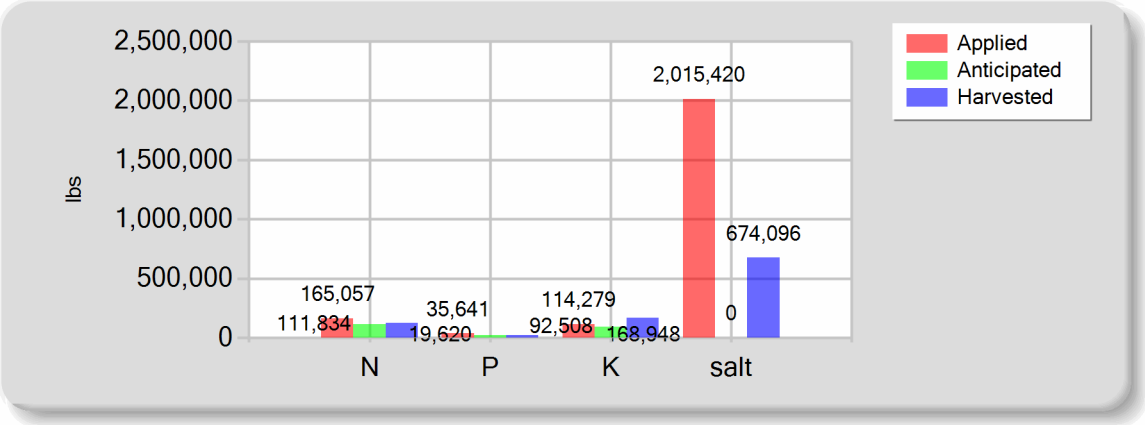
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

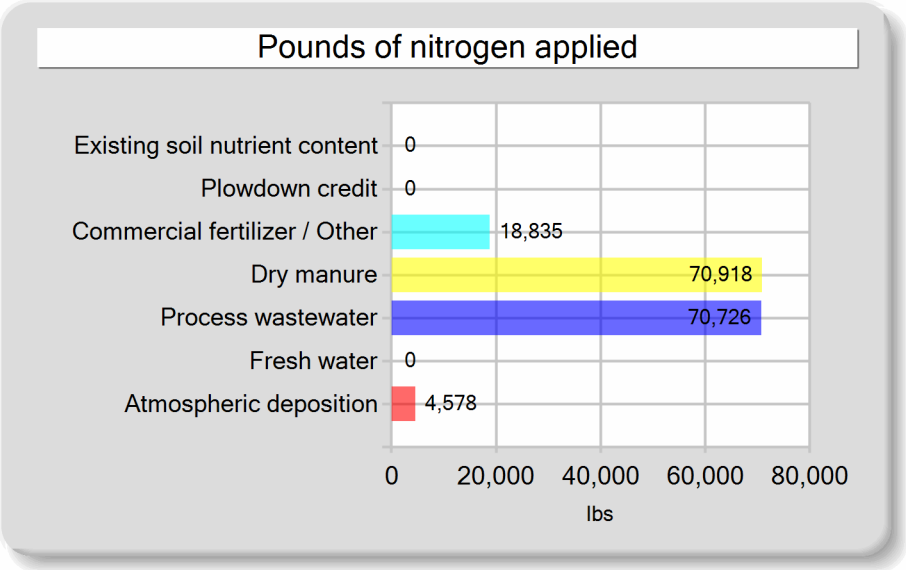
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

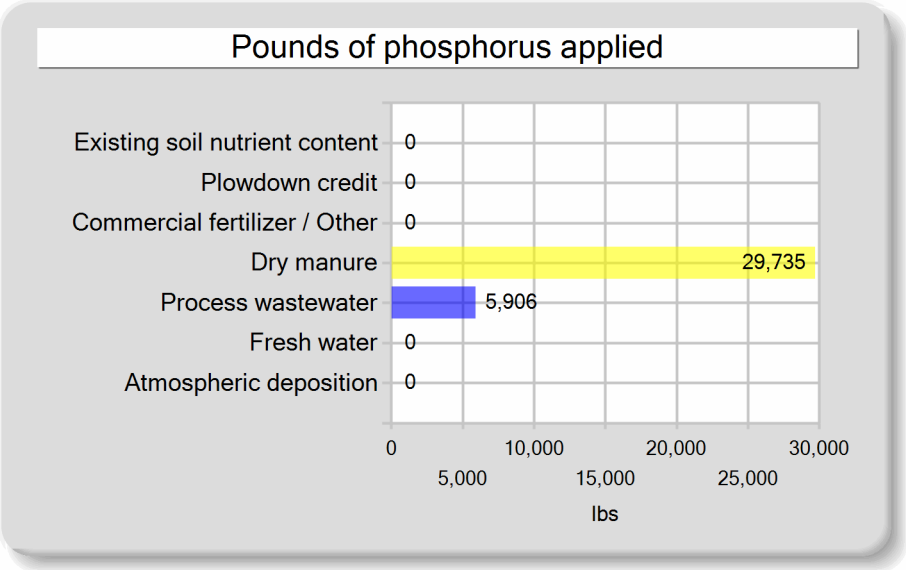
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	18,835.20	0.00	0.00	0.00
Dry manure	70,917.81	29,735.14	42,974.74	796,507.33
Process wastewater	70,726.14	5,905.86	71,304.02	619,433.33
Fresh water	0.00	0.00	0.00	599,479.10
Atmospheric deposition	4,578.00	0.00	0.00	0.00
Total nutrients applied	165,057.16	35,641.00	114,278.76	2,015,419.75
Anticipated crop nutrient removal	111,834.00	19,620.00	92,508.30	0.00
Actual crop nutrient removal	124,720.30	22,989.38	168,948.02	674,096.15
Nutrient balance	40,336.86	12,651.62	-54,669.26	1,341,323.61
Applied to removed ratio	1.32	1.55	0.68	2.99

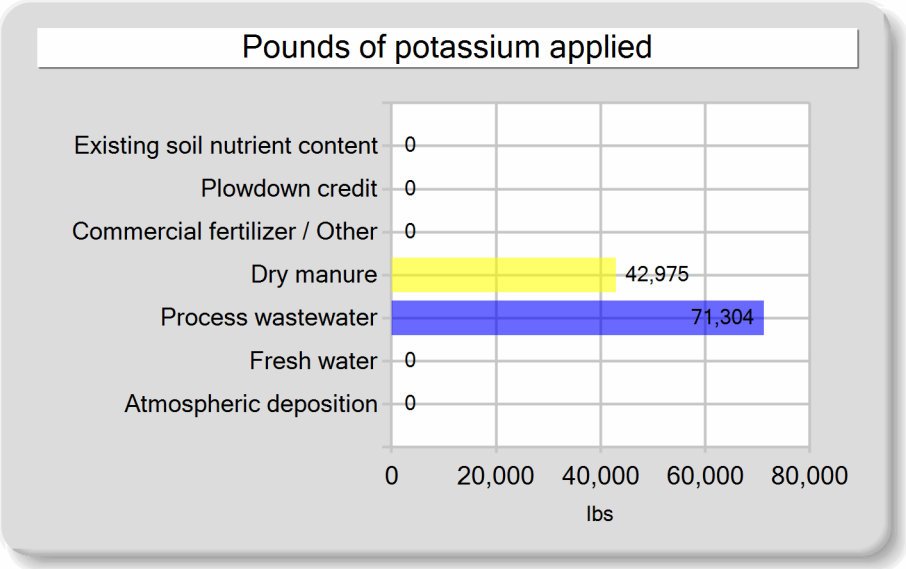
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

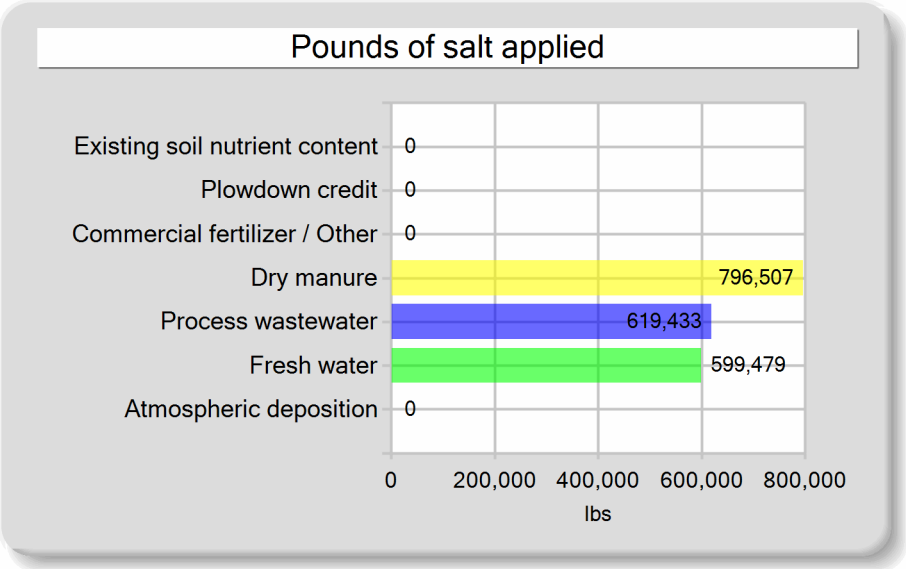


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Corn Home Fields 1 & 2 Corn received a combination of both process wastewater and commercial fertilizer which resulted in field exceeding a 1.6 nitrogen application to removal ratio. Though these applications are in line with the Nutrient Management plan, in the future greater care will be taken in reviewing analytical information prior to applying any additional nitrogen.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

John Toste	John Toste
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
John Toste	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 27, 2024	Jun 27, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: John Toste

Name of Dairy Facility: Toste Dairy

Facility Address: 4609 Santa Fe Grade Rd Newman 95360
Number and Street City Zip Code

Contact Person Name and Phone Number: John Toste 209-862-1812
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: J & F Fertilizer

Address of Hauling Company /Person: 27272 Gun Club Rd Gustine 95322
Number and Street City Zip Code

Contact Person: John Nunes 209-854-6325
Name Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Mike Robinson 909 Real Ave Newman 95360 209-862-3833
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

054-090-003
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: 4/21/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 2920 Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): 72.4% / 61.6% Moisture
Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: Number of loads multiplied by load weight

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: John Tosta Date: Jun 27, 2024

Hauler's Signature: John Nunes Date: Jun 27, 2024

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
John Toste Dairy
785 Santa Fe Grade Rd.
Newman, CA 95360

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 9/27/23 10:30 AM
SUBMITTED BY: Sandra
ANALYZED DATE: 9/27/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 10.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW-Dairy #1	T3270032A	Electrical Conductivity (EC)	EPA 120.1 and	1.25	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	5.58	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	801	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.4	93.4 (T3269010A)	95.8 (T3269010A)	2.53
Electrical Conductivity (EC)	97.9			0.208
Nitrate Nitrogen (NO3-N)	98.2	98.9 (T3269039A)	91.9 (T3269039A)	7.42

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ELAP Certificate No.: 2714

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Compliance Analysis Report

Customer
John Toste Dairy
785 Santa Fe Grade Rd.
Newman, CA 95360

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 9/27/23 10:30 AM
SUBMITTED BY: Sandra
ANALYZED DATE: 9/27/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 11.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW-Dairy #2	T3270032B	Electrical Conductivity (EC)	EPA 120.1 and	1.28	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	5.63	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	820	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.4	93.4 (T3269010A)	95.8 (T3269010A)	2.53
Electrical Conductivity (EC)	97.9			0.208
Nitrate Nitrogen (NO3-N)	98.2	98.9 (T3269039A)	91.9 (T3269039A)	7.42

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Compliance Analysis Report

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785 Santa Fe Grade Rd.
Newman, CA 95360

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 9/27/23 10:30 AM
SUBMITTED BY: Sandra
ANALYZED DATE: 9/27/23

Irrigation /Ground Water (H14)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 11.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW-Silva Ranch	T3270032C	Electrical Conductivity (EC)	EPA 120.1 and	1.69	mmhos/cm
		Bicarbonate Alkalinity (ALK-B)	SM 2320B-2011	292	mg/L
		Carbonate Alkalinity (Carb)	SM 2320B-2011	< 2.00	mg/L
		Sulfate (SO4)	EPA 375.2	21.4	mg/L
		Sodium (Na)	EPA 200.7	151	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Magnesium (Mg)	EPA 200.7	53.3	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	7.42	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	960	mg/L
		Calcium (Ca)	EPA 200.7	98.8	mg/L
		Chloride (Cl)	SM 4500-Cl D-2011	34.9	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1080	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	96.6	94.6 (T3268021A)	96.4 (T3268021A)			1.88
Calcium (Ca)	91.0			144 (T3268020A)	144 (DUP)	0.014
Chloride (Cl)	105	104 (T3277024A)	106 (T3277024A)			1.20
Magnesium (Mg)	92.0			19.7 (T3268020A)	19.7 (DUP)	0.406
Nitrate Nitrogen (NO3-N)	99.7	102 (T3277024A)	98.3 (T3277024A)			3.46
Sodium (Na)	97.1			330 (T3268020A)	330 (DUP)	0.036
Sulfate (SO4)	102	104 (T3277024A)	102 (T3277024A)			1.58
Total Dissolved Solids (TDS)	96.0			690 (T3275031A)	655 (Dup)	5.20

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Compliance Analysis Report

Customer
John Toste Dairy
785 Santa Fe Grade Rd.
Newman, CA 95360

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 9/27/23 10:30 AM
SUBMITTED BY: Sandra
ANALYZED DATE: 9/27/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 10.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW-Azevedo	T3270032D	Electrical Conductivity (EC)	EPA 120.1 and	2.20	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	13.2	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1410	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.4	93.4 (T3269010A)	95.8 (T3269010A)	2.53
Electrical Conductivity (EC)	97.9			0.208
Nitrate Nitrogen (NO3-N)	98.2	98.9 (T3269039A)	91.9 (T3269039A)	7.42

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Compliance Analysis Report

Customer
John Toste Dairy
785 Santa Fe Grade Rd.
Newman, CA 95360

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 9/27/23 10:30 AM
SUBMITTED BY: Sandra
ANALYZED DATE: 9/27/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 11.5c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW-Costa	T3270032E	Electrical Conductivity (EC)	EPA 120.1 and	0.947	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	4.23	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	606	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.4	93.4 (T3269010A)	95.8 (T3269010A)	2.53
Electrical Conductivity (EC)	97.9			0.208
Nitrate Nitrogen (NO3-N)	98.2	98.9 (T3269039A)	91.9 (T3269039A)	7.42

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Compliance Analysis Report

Customer
John Toste Dairy
785 Santa Fe Grade Rd.
Newman, CA 95360

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 9/27/23 10:30 AM
SUBMITTED BY: Sandra
ANALYZED DATE: 9/27/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 11.4c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW-Botelho	T3270032F	Electrical Conductivity (EC)	EPA 120.1 and	1.53	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	6.47	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	982	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.4	93.4 (T3269010A)	95.8 (T3269010A)	2.53
Electrical Conductivity (EC)	97.9			0.208
Nitrate Nitrogen (NO3-N)	98.2	98.9 (T3269039A)	91.9 (T3269039A)	7.42

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Compliance Analysis Report

Customer
John Toste Dairy
785 Santa Fe Grade Rd.
Newman, CA 95360

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 9/27/23 10:30 AM
SUBMITTED BY: Sandra
ANALYZED DATE: 9/27/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 11.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW-House	T3270032G	Electrical Conductivity (EC)	EPA 120.1 and	0.802	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	4.48	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	513	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.4	93.4 (T3269010A)	95.8 (T3269010A)	2.53
Electrical Conductivity (EC)	97.9			0.208
Nitrate Nitrogen (NO3-N)	98.2	98.9 (T3269039A)	91.9 (T3269039A)	7.42

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DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: John Tosk
Mailing address:
(not facility address) Newman
Phone#: 209-862-1513

Authorized Copy Released To: F & R
Sampled By:
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	Analysis: H1	H3	M3	F18	OTHER
1	9-26-23	5:00pm	DW - Dairy #1		10.6'		X				
2		5:00pm	DW - Dairy # 2		11.2'		X				
3		3:00pm	DW - Silva Ranch		11.2'						H14
4		1:00pm	DW - Azevedo		10.8'		X				
5		12:30pm	DW - Costa		11.2'		X				
6		12:00pm	DW - Botelho		11.4'		X				
7		11:30AM	DW - House		11.0'		X				
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
1st	Sandal					
2nd		DENELE ANALYTICAL, INC.	9-27-23 10:30			T3270032

Required to be submitted to any agency? (YES / NO)

DELLAVALLE™
LABORATORY INC

F & R Ag Services
2857 Geer Rd Ste A
Turlock, CA 95382

Account# 00-0021443
Account Manager: Ben Nydam
Submitted By: F&R Ag Services
Ranch: CCID

Received: 07/25/2023 16:26
Reported: 07/31/2023 11:05

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23G1807-01	CCID	Ag Water	Sean Locke		07/25/2023 7:30

Default Cooler Temperature on Receipt °C: 2.9
Containers Intact
COC/Labels Agree
Preservation Confirmed
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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DELLAVALLE™
LABORATORY INC

F & R Ag Services
2857 Geer Rd Ste A
Turlock, CA 95382

Account# 00-0021443
Account Manager: Ben Nydam
Submitted By: F&R Ag Services
Ranch: CCID

Received: 07/25/2023 16:26
Reported: 07/31/2023 11:05

Sample Results

Sample: CCID
23G1807-01 (Water)

Sampled: 7/25/2023 7:30
Sampled By: Sean Locke

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.23**	mmhos/cm	0.01	1		07/28/23 11:23	SM 2510 B		BEG0956
Electrical Conductivity umhos	**228**	umhos/cm	10.0	1		07/28/23 11:23	SM 2510 B		BEG0956
Nitrate Nitrogen as NO3N	**0.3**	mg/L	0.1	1	10	07/26/23 01:15	EPA 300.0		BEG0856
pH	**6.7**	units	1.0	1		07/28/23 11:23	SM 4500-H+	H	BEG0956
Total Filterable Solids (TDS)	**155**	mg/L	10.0	1		07/28/23 15:53	SM 2540 C		BEG0941
Temperature	**25.0**	°C	0.0	1		07/28/23 11:23	SM 2510 B		BEG0956
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		07/28/23 15:34	SM 4500-NH3 C		BEG1051
Total Nitrogen	ND	mg/L	1.00	1		07/28/23 15:34	SM 4500-NH3 C		BEG1051

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Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEG0856										
Blank (BEG0856-BLK1)					Prepared & Analyzed: 7/25/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEG0856-BLK2)					Prepared & Analyzed: 7/25/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEG0856-BLK3)					Prepared & Analyzed: 7/26/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEG0856-BS1)					Prepared & Analyzed: 7/25/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		102	90-110		
LCS (BEG0856-BS2)					Prepared & Analyzed: 7/26/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		104	90-110		
Duplicate (BEG0856-DUP1)		**Source: 23G1760-01**			Prepared & Analyzed: 7/25/2023					
Nitrate Nitrogen as NO3N	1.7		0.1	mg/L		1.7			0.0592	10
Duplicate (BEG0856-DUP2)		**Source: 23G1807-01**			Prepared & Analyzed: 7/26/2023					
Nitrate Nitrogen as NO3N	0.3		0.1	mg/L		0.3			0.669	10
Matrix Spike (BEG0856-MS1)		**Source: 23G1760-01**			Prepared & Analyzed: 7/25/2023					
Nitrate Nitrogen as NO3N	6.9		0.1	mg/L	5.000	1.7	105	90-110		
Matrix Spike (BEG0856-MS2)		**Source: 23G1807-01**			Prepared & Analyzed: 7/26/2023					
Nitrate Nitrogen as NO3N	5.4		0.1	mg/L	5.000	0.3	101	90-110		
Reference (BEG0856-SRM1)					Prepared & Analyzed: 7/25/2023					
Nitrate Nitrogen as NO3N	10.1			mg/L	10.00		101	90-110		
Reference (BEG0856-SRM2)					Prepared & Analyzed: 7/25/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEG0856-SRM3)					Prepared & Analyzed: 7/26/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		

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LABORATORY INC

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2857 Geer Rd Ste A
Turlock, CA 95382

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Account Manager: Ben Nydam
Submitted By: F&R Ag Services
Ranch: CCID

Received: 07/25/2023 16:26
Reported: 07/31/2023 11:05

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEG0941										
Blank (BEG0941-BLK1)					Prepared: 7/25/2023	Analyzed: 7/28/2023				
Total Filterable Solids (TDS)	ND		10.0	mg/L						
LCS (BEG0941-BS1)					Prepared: 7/25/2023	Analyzed: 7/28/2023				
Total Filterable Solids (TDS)	37.5		10.0	mg/L	2000		1.88	0-200		
Duplicate (BEG0941-DUP1)		**Source: 23G1523-01**			Prepared: 7/25/2023	Analyzed: 7/28/2023				
Total Filterable Solids (TDS)	2830		10.0	mg/L		2750			2.87	10
Duplicate (BEG0941-DUP2)		**Source: 23G1819-02**			Prepared: 7/25/2023	Analyzed: 7/28/2023				
Total Filterable Solids (TDS)	313		10.0	mg/L		337			7.18	10
Reference (BEG0941-SRM1)					Prepared: 7/25/2023	Analyzed: 7/28/2023				
Total Filterable Solids (TDS)	353			mg/L	325.0		109	90-110		
Reference (BEG0941-SRM2)					Prepared: 7/25/2023	Analyzed: 7/28/2023				
Total Filterable Solids (TDS)	530			mg/L	495.0		107	90-110		

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Ranch: CCID

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEG0956										
Blank (BEG0956-BLK1)					Prepared: 7/26/2023	Analyzed: 7/28/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	6.2		1.0	units						
Temperature	25.0		0.0	°C						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEG0956-BLK2)					Prepared: 7/26/2023	Analyzed: 7/28/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	7.0		1.0	units						
Temperature	25.0		0.0	°C						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEG0956-BLK3)					Prepared: 7/26/2023	Analyzed: 7/28/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	7.1		1.0	units						
Temperature	25.0		0.0	°C						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Duplicate (BEG0956-DUP1)		**Source: 23G1819-03**			Prepared: 7/26/2023	Analyzed: 7/28/2023				
pH	7.8		1.0	units		7.8			0.641	10
Electrical Conductivity	0.43		0.01	mmhos/cm		0.43			0.0233	10
Electrical Conductivity umhos	429		10.0	umhos/cm		428			0.0233	10
Duplicate (BEG0956-DUP2)		**Source: 23G1821-05**			Prepared: 7/26/2023	Analyzed: 7/28/2023				
Electrical Conductivity	0.55		0.01	mmhos/cm		0.55			0.707	10
pH	7.7		1.0	units		7.6			0.654	10
Electrical Conductivity umhos	550		10.0	umhos/cm		554			0.707	10
Reference (BEG0956-SRM1)					Prepared: 7/26/2023	Analyzed: 7/28/2023				
Electrical Conductivity	581			umhos/cm	538.0		108	90-110		
Reference (BEG0956-SRM2)					Prepared: 7/26/2023	Analyzed: 7/28/2023				
pH	5.8			units	5.820		100	28178-101.71		
Reference (BEG0956-SRM3)					Prepared: 7/26/2023	Analyzed: 7/28/2023				
Electrical Conductivity	1080			umhos/cm	1000		108	90-110		
Electrical Conductivity umhos	1080			umhos/cm	1000		108	90-110		
Reference (BEG0956-SRM4)					Prepared: 7/26/2023	Analyzed: 7/28/2023				
Electrical Conductivity	1080			umhos/cm	1000		108	90-110		
Electrical Conductivity umhos	1080			umhos/cm	1000		108	90-110		
Reference (BEG0956-SRM5)					Prepared: 7/26/2023	Analyzed: 7/28/2023				
Electrical Conductivity	1070			umhos/cm	1000		107	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEG0956 (Continued)										
Reference (BEG0956-SRM5)					Prepared: 7/26/2023	Analyzed: 7/28/2023				
Electrical Conductivity umhos	1070			umhos/cm	1000		107	90-110		
Reference (BEG0956-SRM6)					Prepared: 7/26/2023	Analyzed: 7/28/2023				
pH	4.0			units	4.000		99.2	97.5-102.5		
Reference (BEG0956-SRM7)					Prepared: 7/26/2023	Analyzed: 7/28/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEG0956-SRM8)					Prepared: 7/26/2023	Analyzed: 7/28/2023				
pH	4.0			units	4.000		99.8	97.5-102.5		

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2857 Geer Rd Ste A
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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEG1051										
Blank (BEG1051-BLK1)					Prepared: 7/27/2023	Analyzed: 7/28/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
Blank (BEG1051-BLK2)					Prepared: 7/27/2023	Analyzed: 7/28/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
LCS (BEG1051-BS1)					Prepared: 7/27/2023	Analyzed: 7/28/2023				
Kjeldahl Nitrogen (TKN), Total	5.74		1.00	mg/L	5.709		101	90-110		
LCS (BEG1051-BS2)					Prepared: 7/27/2023	Analyzed: 7/28/2023				
Kjeldahl Nitrogen (TKN), Total	5.92		1.00	mg/L	5.709		104	90-110		
Duplicate (BEG1051-DUP1)		**Source: 23G0034-01**			Prepared: 7/27/2023	Analyzed: 7/28/2023				
Kjeldahl Nitrogen (TKN), Total	16.0		3.50	mg/L		15.1			5.33	10
Duplicate (BEG1051-DUP2)		**Source: 23G1824-01**			Prepared: 7/27/2023	Analyzed: 7/28/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.40	mg/L		ND				10
Matrix Spike (BEG1051-MS1)		**Source: 23G0034-01**			Prepared: 7/27/2023	Analyzed: 7/28/2023				
Kjeldahl Nitrogen (TKN), Total	28.0		3.50	mg/L	9.990	15.1	128	90-110		
Matrix Spike (BEG1051-MS2)		**Source: 23G1824-01**			Prepared: 7/27/2023	Analyzed: 7/28/2023				
Kjeldahl Nitrogen (TKN), Total	7.61		1.40	mg/L	7.992	ND	95.2	90-110		
Reference (BEG1051-SRM1)					Prepared: 7/27/2023	Analyzed: 7/28/2023				
Kjeldahl Nitrogen (TKN), Total	23.5			mg/L	23.80		98.6	90-110		

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07/25/23 16:26 23G1807

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8174

No. of **Water** Samples 1 No. of Bottles 2

7/25/23

Water Type: [ ] Drinking Water [ ] Wastewater
[✓] Ag Water [ ] Groundwater [ ] Monitoring Well
[ ] Supply Water

Other: No. of Samples **Soil** _____ **Tissue** _____ **Manure** _____

Logged By _____

BILL TO 21443/09 Acct#/Cons. _____ Purchase Order No

Results Needed By _____

FtR AG SERVICES
Company Name

2857 GEER RD. SUITE A
Address

TURLOCK, CA 95382
City, State Zip

209-252-1408
Telephone

Fax

Cell/Email

REQUESTED BY F&R Ag Services

PROJECT CCID

Crop _____ Intended _____

Sampled By SEAN LOCKE

Analysis and Bottles Required:

[ ] Well Water **(DWW1)** - EC, NO_3-N, Field NH_4-N, if detected run NH_4-N (1-1 liter plastic, unpreserved) **(Annually)**
[ ] Well Water **(DWW2)** - DWW1 plus HCO_3, CO_3, Cl, SO_4, Dissolved: Ca, Mg, Na, TDS (1-1 liter plastic, unpreserved) **(Every 5 yrs)**
[ ] Process Water **(DPW1)** - EC, NO_3-N, NH_4-N, TKN, TDS, TP, TK (1-1 liter plastic, unpreserved) **(Quarterly)**
[ ] Process Water **(DPW2)** - DPW1 plus CO_3, HCO_3, Cl, SO_4, Total: Ca, Mg, Na, (1-1 liter plastic, unpreserved) **(Every 2 yrs)**
[X] Canal Water **(DCW1)**-EC, NO_3-N,TKN,TN,TDS (1-1 liter plastic,unpreserved)
[ ] Water – EC, NO_3-N, TDS, TP (1-1 liter plastic, unpreserved)
[ ] Manure **(DM1)** - N, P, K, Moisture **(2x/yr)**
[ ] Manure **(DM2)** - DM1 plus Ash, Ca, Mg, Na, S, Cl **(Every 2 yrs)**
[ ] Compost **(OSA2)** – Moisture, N, P, K, Ash, OM, C:N ratio, Ca, Mg, Na, Fe, Cu, Mn, Zn, S
[ ] Harvest Tissue **(CRA1)** - Moisture , N, P, K, Ash
[ ] Tissue – % Moisture
[ ] Fall Soil – **(DS2)** - FA1 (0-1') **plus CEC, OM**; NO_3-N (1-2')
[ ] OTHER:

COPIES TO: labs@fragservices.com, slocke@fragservices.com
[ ] Co Health Dept
[ ] RWQCB [X] Copy of Chain
[ ] State Forms [ ] QA/QC Documents

Description of Sample	Field NH_4-N	Date/Time Sampled	Rec'd Temp °C
1. CCID		7/25 7:30	2.9
2.			
3.			
4.			
5			
6.			
7.			
8.			
9.			
10.			

IR Thermometer SN: 192603727
Correction Factor: 0°C
Calibration Due: 9/26/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First		F+R	7/25 7:30	
Second	LD	DLI	7/25 1:35	7/25 3:40
Third				
Fourth		DLI	7-25-23 16:26	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through *cal* under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Shipping
$_____ In
$_____ Out

Amt Paid _____ Rec By _____ Check # _____ Date _____

Signature _____

Paste Label Here

7·25·23

Shipping Information: Shipped In ☒ Picked-Up ☐ Walk In ☐ DLI Sampler ☑ Other ☐ _____

☐ Samples refridgerated before pick up | ☐ Picked up samples placed in Ice chest

Container: Ice Chest ☐ Box ☐ None ☐ | **Refrigerant:** Wet Ice ☐ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☑ Preserved Upon Receipt at Laboratory

	Type of Container(s) Received	Sample Number 1	2	3	4	5	6	7	8	9	10
	Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
Plastics	250 mL unpreserved (White) Plastic	1									
Plastics	250 mL HNO_3 (Red) Plastic										
Plastics	* pH Value										
Plastics	250 mL H_2SO_4 (Yellow) Plastic										
Plastics	* pH Value	<2									
Plastics	500 mL unpreserved (White) Plastic										
Plastics	1 L unpreserved (White) Plastic										
Plastics	1 L unpreserved (BOD) (Purple) Plastic										
Special	500mL unpreserved (White) Glass										
Special	PO4-P Kit										
Special	**Other:**										
	Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
Plastics	250 mL unpreserved (White) Plastic										
Plastics	250 mL HNO_3 (Red) Plastic										
Plastics	250 mL H_2SO_4 (Yellow) Plastic										
Plastics	500 mL HNO_3 (Red)										
Plastics	1 L unpreserved (White) Plastic										
Plastics	1 L unpreserved (BOD) (Purple) Plastic										
Plastics	1 L HNO_3 (Red)										
VOA Vials	40 mL VOA, $Na_2S_2O_3$ + MCAA (EPA531)										
VOA Vials	40 mL VOA, $Na_2S_2O_3$ (EPA547)										
VOA Vials	40mL AG VOA unpreserved (White) (Set of 3)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
VOA Vials	40mL VOA, H_3PO_4 (Set of 3)										
VOA Vials	40 mL VOA, HCl (Blue) (Set of 3)										
VOA Vials	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
Glass	250 mL AG H_2SO_4 (Yellow)										
Glass	250 mL AG $Na_2S_2O_3$ (Green)										
Glass	250 mL AG $Na_2S_2O_3$ + MCAA										
Glass	500 mL glass unpreserved (White)										
Glass	500 mL AG HCl (Blue)										
Glass	1 L AG unpreserved (White)										
Glass	1 L AG H_2SO_4 (Yellow)										
Glass	1 L AG $Na_2S_2O_3$ (Green)										
Glass	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
Special	Cyanide - 500 mL NaOH										
Special	Asbestos - 1L P wrapped in foil (Set of 2)										
Special	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
Special	Chlorite/Bromate - 250 mL AG with EDA										
Special	HAA5 - 250mL AG Ammonium Chlorite										
Special	DO KIT										
Special	**Other:**										
Special	**Other:**										

pH Strips
Lot 10BDH4501
Exp. Jan. 2025

07/25/23 16:26 23G1807

DRAFT
SEMIDOC

2023 Annual Report John Toste Dairy Final

Final Audit Report 2024-06-27

Created:	2024-06-27
By:	Kierstan Rhodes (krhodes@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAA2EY4Lk21sNhozaExJ1veB7WsdNCKlV0d

"2023 Annual Report John Toste Dairy Final" History

Document created by Kierstan Rhodes (krhodes@fragservices.com)
2024-06-27 - 2:33:45 PM GMT

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